

Notation and equation-to-code crosswalk

This document is the reference for the symbols used in the main text and the Supplementary Information, and for the correspondence between the equations in those documents and the functions in `src/signaling/`. It has three parts: a symbol table grouped by framework layer, a crosswalk from each labelled equation to the function(s) that implement it, and the reverse crosswalk from every public function in `src/signaling/` to the equation or section it serves.

Conventions. "Main" refers to the main text and "Supp." to the Supplementary Information; locations are given by section label (for example `sec:layer2`) or equation label (for example `eq:pconflict`). Calibrated values are quoted at the empirical anchor, the within-group reward $\lambda_W = 0.68$ with the self-consistent cost $C = C_{\text{model}}(\lambda_W) \approx 0.35$, group size $n = 12$, and the $\delta = 1$ reference stock $M_g = l_g \approx 10.3$, unless a row says otherwise. Every default listed in the "Value" column is defined in `src/signaling/calibration.py` under the constant named in the "Code" column; derived anchor values were computed from the current code and agree with the manuscript to the precision quoted. A dash in the "Value" column marks a free variable or sweep axis. Three equations with no standalone implementing function are listed at the end.

Symbol overloads. Four symbols carry more than one meaning in the manuscript, each scoped to its section. γ is the buffering efficiency (Layer 3, 0.3), the concave-benefit exponent in the functional-form robustness check (Supp. `sec:si-functional-form`), and the placement resource-cost coefficient (Supp. `sec:si-placement`). D is the fighting cost in the mutual-assessment escalation threshold (2.0) and the diffusion coefficient of the reaction-diffusion equation (1 in all figures). v is the normalized contest value in the escalation threshold (1.0), the contested stake in the war-of-attrition deterrence derivation, and the prize in the Tullock contest (1.0). τ is the depreciation time constant $-1/\ln(1 - \delta)$ and the exponential-saturation scale of the alternative network form. α denotes the vulnerability coefficient $\alpha(k)$ and, locally, the slope of the piecewise-linear network form. T is the escalation threshold; the feedback map of Supp. `sec:si-banach` is written $\mathcal{T}$ to keep them distinct. r is the war-avoidance reduction; the Pearson correlation in Supp. `sec:si-network` is a different quantity.

1. Symbol table

1.1 Layer 1: individual signaling equilibrium

Symbol	Meaning	Units / range	Value	Manuscript	Code
i	Individual contributor (the person); each act is a Spence signal and a credibility-enhancing display	index	person	Main <code>tab:notation</code>	<code>layer1</code>
q, q_i	Individual productive capacity, the latent trait the signal reveals	payoff units, $q > 0$	-	Main Layer 1	<code>q</code> arguments in <code>layer1</code>
$q_{\min}, q_{\max}$	Capacity floor and ceiling of the uniform capacity distribution	payoff units	0.1, 2.0	Supp. <code>tab:params</code>	<code>DEFAULT_Q_MIN, DEFAULT_Q_MAX</code>
$E[q]$	Mean capacity under the uniform distribution	payoff units	1.05	Supp. <code>eq:sw</code>	derived
x, x_i	Labor invested in monument construction	payoff units, $x \geq 0$	-	Main Layer 1	<code>x</code> arguments
$c(x, q)$	Construction cost $x^2/(2q)$	payoff units	-	Main <code>eq:spence</code>	<code>layer1.quadratic_cost</code>
$\partial^2 c / \partial x \partial q$	Spence (single-crossing) cross-derivative, $-x/q^2 < 0$	-	-	Main <code>eq:spence</code>	<code>layer1.verify_spence_condition</code>
$\hat{q}(x)$		payoff units	-	Main Layer 1	<code>layer1.receiver_inference</code>

Symbol	Meaning	Units / range	Value	Manuscript	Code
	Capacity observers infer from investment x ; $\hat{q}(x^*(q)) = q$ at equilibrium				
$b(x)$	Signaling benefit $\lambda_W \hat{q}(x)$	payoff units	-	Main Layer 1	layer1.linear_benefit
λ_W	Within-group positional reward; the sole reward pricing the schedule	payoff units per unit inferred capacity	0.68 (anchor); 0.30 (emergence stage); 0.5 (exploration)	Main tab:notation	DEFAULT_LAMBDA_W_ANCHOR, DEFAULT_LAM_W, DEFAULT_LAMBDA
$x^*(q)$	Separating-equilibrium schedule $\sqrt{(\lambda_W(q^2 - q_{\min}^2))}$	payoff units	-	Main eq:xstar	layer1.equilibrium_investment
$w^*(q)$	Equilibrium payoff $q(1 + \lambda_W/2) + \lambda_W q_{\min}^2/(2q)$	payoff units	-	Main fig:equilibrium; Supp. sec:si-fitness-gain	layer1.equilibrium_fitness
$\Delta w(q)$	Equilibrium-vs-baseline diagnostic $\lambda_W q/2 + \lambda_W q_{\min}^2/(2q)$ (not the free-rider resolution)	payoff units	-	Supp. eq:deltaw	layer1.fitness_gain
$\Delta_{fr}(q)$	Free-rider deterrent $\lambda_W(q - q_{\min})^2/(2q) \geq 0$	payoff units	-	Supp. eq:freerider_deterrent	layer1.free_rider_deterrent
μ_N, μ_{N^*}	Perceived non-builder capacity and the cutoff $E[q] - C_{\text{model}}/\lambda_W$ above which the mean deterrent turns negative	payoff units	$\mu_N = q_{\min}$ (headline); $\mu_{N^*} \approx 0.53$	Supp. sec:si-free-rider	layer1.participation_equilibrium
$C_{\text{model}}(\lambda_W)$	Average per-capita equilibrium cost $E[c(x^*(q), q)]$ under uniform capacity	fraction of baseline fitness	≈ 0.352 at $\lambda_W = 0.68$	Supp. eq:Cmodel	layer1.average_equilibrium_cost
C	Reproductive cost of building entering the threshold and β_0 ; equals C_{model} on the self-consistent locus	fraction	≈ 0.35 ; bracket [0.20, 0.50]	Main tab:notation	RAPA_NUI["C"], C_exogenous arguments
C_{emp}	Architectural-energetics anchor for C (per-builder peak labor fraction)	fraction	≈ 0.35 ; $\lambda_W \approx 1.93$ C_{emp}	Supp. sec:si-Cmodel	layer1.lambda_W_for_C; CALIBRATION_TARGETS["labor_diversion"]
$B(\lambda_W)$	Average of the diagnostic Δw over uniform capacity	payoff units	≈ 0.362 at anchor	Supp. eq:Blambda	price_equation.average_signaling_benefit
s_W	Within-group positional differential $\lambda_W(E[q] - q_{\min})$	fraction	≈ 0.646	Supp. eq:sw	price_equation.within_group_status_differential
n	Size of the face-to-face cohabiting community	persons	sweep [10, 250]; illustration 12	Main tab:notation	DEFAULT_N_COMPUTE
I_g	Per-period group investment flow $\sum_i x^*(q_i) = n E[x^*(q)]$	payoff units	≈ 10.3	Main tab:notation	layer1.group_monument_stock, layer1.expected_monument_stock
$U(q', q)$		payoff units	-		layer1.fitness_at_deviation

Symbol	Meaning	Units / range	Value	Manuscript	Code
	Payoff of reporting q' when true capacity is q (global incentive compatibility)			Supp. sec:si-functional-form	
a, b (cost)	Power-cost exponents in $c = x^a / (a q^b)$; quadratic is $a = 2, b = 1$	-	robustness check uses $a = 2.5, b = 1.5$	Supp. sec:si-functional-form	layer1.power_cost

1.2 Channel selection

Symbol	Meaning	Units / range	Value	Manuscript	Code
s	Channel index: monument (M), feast (F), ritual (R), hunting (H), verbal (V)	-	-	Supp. sec:si-channel	layer1.SignalChannel, layer1.ALL_CHANNELS
a_s^W, a_s^C, a_s^X	Audience activation weights of channel s for the within-group, competitive, and cooperative audiences	[0, 1]	monument (1, 1, 1); feast (0.9, 0.2, 0.2); hunting (0.6, 0.05, 0.15); ritual (0.7, 0.1, 0.1); verbal (0.3, 0, 0)	Supp. sec:si-channel-decomp	layer1.MONUMENT_CHANNEL and siblings
$\tilde{\lambda}_C, \tilde{\lambda}_X$	Emergence-stage audience-level between-group returns (distinct from the derived marginal returns λ_C, λ_X)	payoff units	0.15, 0.15	Supp. tab:params	DEFAULT_LAM_C, DEFAULT_LAM_X
λ_s	Channel effective reward $a_s^W \lambda_W + a_s^C \tilde{\lambda}_C + a_s^X \tilde{\lambda}_X$	payoff units	-	Supp. sec:si-channel-decomp	layer1.channel_effective_lambda, layer1.lambda_decomposition
ρ_s	Signal fidelity of channel s	[0, 1]	ρ_M 0.95, ρ_F 0.50, ρ_H 0.40, ρ_R 0.30, ρ_V 0.05	Supp. tab:params	DEFAULT_RHO_MONUMENT and siblings; layer1.signal_fidelity
windfall probability	Fraction of high-signal feasts funded by transient windfall; degrades ρ_F	[0, 1]	0.3	Supp. sec:si-channel	DEFAULT_WINDFALL_PROB
κ_s	Channel marginal-cost factor in $c_s = \kappa_s x^2 / (2q)$; cancels from the equilibrium net payoff	-	1 for the single-channel model	Supp. sec:si-channel	not represented in code, because it cancels: channel_dominance_condition uses $\lambda_s \rho_s$ directly
$x_s^*(q)$	Channel-specific schedule $\sqrt{(\lambda_s \rho_s (q^2 - q_{\min}^2) / \kappa_s)}$	payoff units	-	Supp. sec:si-channel	layer1.equilibrium_investment with $\lambda = \lambda_s \rho_s$
$A(q), \bar{A}$	Payoff multiplier $(q^2 + q_{\min}^2) / (2q)$ and its community mean $E[A(q)]$	payoff units	$\bar{A} \approx 0.533$	Supp. sec:si-channel	layer1.mean_payoff_multiplier
f_s, f_M	Fixed (access) cost of channel s ; f_M is uncalibrated and swept	payoff units	f_M 0.16 (illustrative; swept [0, 0.24]); f 0.011 (feast, ritual, hunting); 0 (verbal)	Supp. tab:params	DEFAULT_FIXED_COST_MONUMENT and siblings
$\bar{V}_s$	Community-mean net return $\lambda_s \rho_s \bar{A} - f_s$; the channel norm settles on $\arg \max_s \bar{V}_s$	payoff units	$\bar{V}_M$ 0.184 vs $\bar{V}_F$ 0.152 at $\lambda_W = 0.68$ with between-group returns zero	Supp. sec:si-channel	layer1.channel_dominance_condition
incremental λ			-		layer1.incremental_lambda

Symbol	Meaning	Units / range	Value	Manuscript	Code
	$\bar{V}_{\text{dominant}} - \bar{V}_{\text{second}}$ (descriptive diagnostic, not a schedule input)	payoff units		Supp. sec:si-channel	

1.3 Signal depreciation

Symbol	Meaning	Units / range	Value	Manuscript	Code
δ	Informational depreciation rate of the signal per period	(0, 1]	0.10 default; $\delta = 1$ is the flow-assessment reference	Main eq:depreciation; tab:notation	DEFAULT_DELTA
$M_g(t)$	Effective monument stock, $(1 - \delta)M_g(t) + I_g(t)$	payoff units	-	Main eq:depreciation	layer1.monument_stock_trajectory
M_g^*	Steady-state stock I_g/δ	payoff units	≈ 10.3 at $\delta = 1$; ≈ 103 at $\delta = 0.10$	Main sec:depreciation	layer1.effective_monument_stock
M_g	Group stock as read by Layers 2 and 3; reference stock is the $\delta = 1$ steady state	payoff units	≈ 10.3	Main tab:notation	layer1.expected_monument_stock
τ	Time constant $-1/\ln(1 - \delta)$ of convergence to M_g^* ($\approx 1/\delta$ for small δ)	periods	-	Main sec:depreciation-results	related: layer1.signal_half_life gives $\ln 2/(-\ln(1 - \delta)) \approx 6.58$ at $\delta = 0.10$
$c_{\text{maint}}(M_g, q)$	Cost $(\delta M_g)^2/(2q)$ of holding the stock against depreciation	payoff units	-	Supp. eq:multiperiod_spence	layer1.maintenance_cost
$\partial^2 c_{\text{maint}}/\partial M_g \partial q$	Multi-period Spence cross-derivative $-\delta^2 M_g/q^2$, zero at $\delta = 0$	-	-	Supp. eq:multiperiod_spence	layer1.verify_multi_period_spence_condition
$R(M_g)$	Renovation-to-new-construction ratio $\delta M_g/(I_g - \delta M_g)$ on $0 \leq M_g < M_g^*$	-	-	Supp. eq:reno_ratio	layer1.maintenance_to_new_ratio

1.4 Layer 2: intergroup assessment and conflict

Symbol	Meaning	Units / range	Value	Manuscript	Code
M_h	Neighboring group's monument stock	payoff units	-	Main sec:layer2	M_h arguments in layer2
RHP	Resource-holding potential (fighting capacity), indexed by M	-	-	Main sec:layer2	-
$\Delta \hat{R}$		RHP units	-	Main sec:layer2	inside layer2.mutual_assessment_conflict_prob

Symbol	Meaning	Units / range	Value	Manuscript	Code
	Perceived RHP difference, $N(M_g - M_h, 2\sigma_{\text{eff}}^2)$				
d	Stock asymmetry $ M_g - M_h $	payoff units	-	Supp. eq:pconflict	inside mutual_assessment_conflict_prob
σ_0	Baseline assessment noise (unrelated to the uncertainty σ)	RHP units	1.0 assumed; decomposition composes to ≈ 0.89	Supp. tab:params; sec:si-noise	DEFAULT_SIGMA_0; layer2.constrained_sigma_0
$\sigma_{\text{eff}}(M_g, M_h)$	Effective noise $\sigma_0/\sqrt{1 + \kappa(M_g + M_h)}$	RHP units	$\leq \sigma_0$	Main sec:layer2	layer2.effective_noise
κ	Information-gain rate of assessment in total stock	per unit stock	0.1	Supp. tab:params	DEFAULT_KAPPA
$\sigma_{\text{deception}}^2$, σ_{lag}^2 , $\sigma_{\text{mismatch}}^2$	Variance components of σ_0^2 : $(1 - \rho_M)V^2$, $\sigma_q^2(1 - \delta)^2/(2\delta - \delta^2)$, $(1 - \rho_{pf})\text{Var}(\text{RHP})$	RHP units ²	0.05, 0.384, 0.36 at defaults	Supp. eq:noise-decomp	layer2.constrained_sigma_0
σ_q	Per-period capacity innovation s.d. (lag component)	RHP units	0.3	Supp. sec:si-noise	DEFAULT_SIGMA_Q
ρ_{pf}	Correlation between productive and fighting capacity	[0, 1]	0.8	Supp. sec:si-noise	DEFAULT_RHO_PROD_FIGHT
Var(RHP)	Variance of fighting capacity across groups	RHP units ²	1.0	Supp. sec:si-noise	DEFAULT_VAR_RHP
T	Escalation threshold (half-width of the perceived-parity band), $T = T_0 V/D$	RHP units	0.25	Supp. eq:pconflict; Supp. tab:params	composed inside layer2.mutual_assessment_conflict_prob
T_0, V, D	Threshold scale, normalized contest value, fighting cost ($D > V$)	-	0.5, 1.0, 2.0	Supp. tab:params	DEFAULT_T_0, DEFAULT_V, DEFAULT_D
Φ	Standard normal cumulative distribution function	-	-	Supp. eq:pconflict	scipy.stats.norm.cdf
β_d	Absolute-deterrence coefficient c/s (per-unit war cost over the stake scale)	per unit stock	0.10	Main tab:notation; Supp. eq:woa-deterrence	DEFAULT_BETA_CONFLICT (argument beta)
P_base	Baseline conflict probability at zero investment	per period	0.01	Supp. tab:params	DEFAULT_P_BASE
P_conflict(M_g, M_h)	Mutual-assessment conflict probability (assessment	probability	$< P_{\text{base}}$ at all positive stock	Supp. eq:pconflict	layer2.mutual_assessment_conflict_prob

Symbol	Meaning	Units / range	Value	Manuscript	Code
	factor times deterrence factor, capped at 1)				
$P_{\text{conflict}}^{\text{self}}$	Self-assessment alternative with escalation propensity $p(M)$	probability	$\approx 0.32, 0.23, 0.14$ at $M = 5, 10, 20$	Supp. eq:pconflict_self	layer2.self_assessment_conflict_prob
$p(M)$	Per-group escalation propensity $1/(1 + \exp(-(M - M_{\text{thresh}})/M_{\text{scale}}))$	probability	-	Supp. eq:pconflict_self	inside self_assessment_conflict_prob
$M_{\text{thresh}}, M_{\text{scale}}$	Self-assessment escalation threshold and sigmoid scale	payoff units	2.0, 1.0	Supp. tab:params	DEFAULT_M_THRESH, DEFAULT_M_SCALE
M_{own}	Bourgeois-convention ownership threshold	payoff units	2.0	Supp. sec:si-self-assessment-sensitivity	DEFAULT_M_OWN, DEFAULT_STEEPNESS
r, r_{bb}	Derived war-avoidance reduction $1 - P_{\text{conflict}}(M, M)/P_{\text{base}}$ on a build-build dyad	fraction	≈ 0.43 at the reference stock; ≈ 0.81 at the $\delta = 0.10$ stock; was assumed 0.75	Main sec:layer2; tab:notation	layer2.derived_conflict_reduction
r_{self}	Self-assessment counterpart $1 - P_{\text{conflict}}^{\text{self}}/P_{\text{base}}$	≤ 1 , may be far below -1	≈ -22 at $M = 10$	Supp. sec:si-self-assessment-sensitivity	layer2.derived_conflict_reduction_self_assessment
α (assessment)	Decay rate of the exponential approximation to the Gaussian assessment factor (robustness variant only)	-	1.0	not in manuscript	DEFAULT_ALPHA_CONFLICT; layer2.mutual_assessment_exponential
c_{rate}	War-of-attrition cost accumulation rate (remark only, not load-bearing)	per period	0.1	Supp. sec:si-woa	DEFAULT_C_RATE; layer2.war_of_attrition_cost
$W(\sigma)$	Expected per-period war cost of an unassessed group, $w_0 \sigma$	fraction of baseline fitness	0.30σ	Main sec:layer2-waravoidance; Supp. sec:si-waravoidance	DEFAULT_WAR_COST
w_0	War-cost coefficient (encounter frequency times per-conflict loss, folded)	fraction	0.30	Supp. tab:params	DEFAULT_WAR_COST
$K(\sigma)$	War-avoidance factor $1 - W(\sigma)(1 - \phi r)$; builder among fully building neighbors $1 - W(1 - r)$, non-builder $1 - W$	fraction	-	Supp. eq:K-waravoidance	inline in price_equation.group_fitness_signaler, group_fitness_nonsignaler, emergence.rare_builder_fitness_advantage
$W(\sigma)\phi r$	War-avoidance advantage $K_{\text{build}} - K_{\text{non}}$	fraction	≈ 0.06 at $\sigma \approx \sigma^*$, $\phi = 1$	Supp. eq:waravoid-advantage	derived from the above

Symbol	Meaning	Units / range	Value	Manuscript	Code
λ_C	Derived marginal competitive return $-dL/dM$ on the group stock (diagnostic; never enters the schedule)	payoff units per unit stock	$\approx 1 \times 10^{-5}$ at the anchor stock ($< 1\%$ of λ_W)	Main tab:notation; Supp. sec:si-banach	layer2.compute_lambda_C
n_nb, f_d, c_w	Neighbor count, dispute frequency, and conflict fitness cost entering only λ_C	-, per pair-period, fraction	4, 0.05, 0.5	Supp. sec:si-banach	DEFAULT_N_NEIGHBORS, DEFAULT_DISPUTE_FREQ, DEFAULT_CONFLICT_COST
m	Conflict mortality of the assumed-coefficient baseline; does not enter the war-avoidance formulation	fraction	0.1	Supp. sec:si-waravoidance	DEFAULT_CONFLICT_MORTALITY

1.5 Layer 3: cooperation networks and crisis buffering

Symbol	Meaning	Units / range	Value	Manuscript	Code
k, k(M_g)	Exchange-partner count (network degree), $k_0 + k_{\max} M_g/(M_{1/2} + M_g)$	partners	≈ 6.70 at the anchor stock	Supp. eq:kdegree	layer3.network_degree
dk/dM_g	Partners gained per unit stock, $k_{\max} M_{1/2}/(M_{1/2} + M_g)^2$	partners per unit stock	-	Supp. eq:lambdaX	layer3.network_degree_derivative
k_0	Baseline degree from kinship and proximity	partners	0.5	Supp. tab:params	DEFAULT_K_0
k_max	Maximum additional partners from signaling	partners	8.0	Supp. tab:params	DEFAULT_K_MAX
M_1/2	Half-saturation stock	payoff units	3.0	Supp. tab:params	DEFAULT_M_HALF
τ (network), K_H, α (slope)	Parameters of the exponential, Hill ($n = 2$), and piecewise-linear alternative forms, each calibrated to match the reference at $M_e \approx 10.3$	-	$\approx 6.92, \approx 5.56, \approx 0.601$	Supp. sec:si-network-robustness	layer3.calibrate_alternative_network_forms, NetworkDegreeSpec
γ	Buffering efficiency per exchange partner	per partner	0.3	Supp. tab:params	DEFAULT_GAMMA
σ	Environmental uncertainty: per-period rate of shortfalls a group cannot weather alone unbuffered; the driving axis	[0, 1]	-	Main sec:layer3; tab:sigma	sigma arguments

Symbol	Meaning	Units / range	Value	Manuscript	Code
$S(\sigma, k)$	Crisis-survival probability $1 - \sigma/(1 + \gamma k)$	probability	-	Main eq:survival	layer3.survival_probability
$\alpha(k)$	Vulnerability coefficient $1/(1 + \gamma k)$, so $S = 1 - \alpha(k)\sigma$	fraction	-	Main sec:layer3	layer3.vulnerability_coefficient
α_{eff}	Derived builder vulnerability $\alpha(k(M_g))$	fraction	≈ 0.33 (reference stock); ≈ 0.29 at $\delta = 0.10$; was assumed 0.30	Main tab:notation	layer3.derive_vulnerability_differential
β_{eff}	Derived non-builder vulnerability $\alpha(k_0)$	fraction	≈ 0.87 ; was assumed 0.90	Main tab:notation	same
λ_X	Derived marginal cooperative return $(dk/dM_g)(\sigma\gamma/(1 + \gamma k)^2)$	payoff units per unit stock	≈ 0.002 at the anchor stock and $\sigma = 0.48$ ($< 1\%$ of λ_W); passes through 0.15 at emergence-stage stocks	Supp. eq:lambdaX	layer3.compute_lambda_X
ρ (network fidelity), p_{connect}	Fidelity in the agent-based signal-read network and edge probability of the baseline random graph	[0, 1]	0.5, 0.15	Supp. sec:si-network	layer3.form_network_signal_based, form_network_baseline; DEFAULT_P_CONNECT_BASE
$\mathcal{T}(\lambda)$	Composite-reward feedback map $\lambda_W + \lambda_C(M(\lambda)) + \lambda_X(M(\lambda); \sigma)$ (robustness variant)	payoff units	fixed point ≈ 0.682 at anchor; $ \mathcal{T}' \leq 0.025$ on the calibrated box	Supp. sec:si-banach	layer3.lambda_total_at_sigma, price_equation.lipschitz_derivative_analytical

1.6 Assembly, threshold, and selection partition

Symbol	Meaning	Units / range	Value	Manuscript	Code
p_g, p	Fraction of individuals within a group who build	[0, 1]	-	Main sec:within-between	p, p_g arguments in price_equation
$\bar{p}$	Population mean of p_g	[0, 1]	-	Main eq:price	price_equation.price_partition
w_S, w_N	Signaler and free-rider fitness, $SK[(1 - C) + s_W(1 - p_g)]$ and $SK[1 - s_W p_g]$	fraction of baseline	-	Supp. eq:full_fitness, eq:member_freerider, eq:wS-si, eq:wN-si	price_equation.within_group_regression (difference), mixed_group_mean_fitness (mean)
W_g	Group mean fitness $(1 - p_g C)SK$; the positional terms cancel	fraction	-	Main sec:within-between	price_equation.mixed_group_mean_fitness
b_i	Peer-excluded positional payoff $\lambda_W(\hat{q}_i - \text{mean of the other } n - 1 \text{ members' } \hat{q})$; sums to zero, own-derivative λ_W	payoff units	-	Supp. eq:positional-game	mean-field form in within_group_regression (no finite-n implementation)
β_0		fraction			

Symbol	Meaning	Units / range	Value	Manuscript	Code
	Within-group selection differential $w_S - w_N = SK(s_W - C) > 0$		> 0 ; $s_W - C \approx 0.29$ at anchor	Main tab:notation; Supp. eq:beta0-si	price_equation.within_group_regression, within_group_selection
$\beta_0/(SK)$	Positional margin $s_W - C = \lambda_W E[(q - q_{\min})^2/(2q)]$	fraction	≈ 0.29 (uniform); ≈ 0.43 (Beta(5,2))	Supp. sec:si-free-rider	layer1.participation_differential_sampled
β_1	Between-group selection differential dW_g/dp	fraction	+0.96 at $p = 0.05$, 0 at p_{opt} , -0.24 at $p = 0.95$ ($\sigma = 0.477$)	Supp. eq:emergence-criterion	price_equation.between_group_regression
F	Assortment of builders into groups	[0, 1]	-	Main sec:novel	F argument
$F\beta_1 + (1 - F)\beta_0$	Selection criterion under assortment	fraction	> 0 for all F at $p \rightarrow 0$	Supp. eq:emergence-criterion	price_equation.emergence_criterion
p_{opt}	Builder fraction maximizing W_g	[0, 1]	≈ 0.24 at $\sigma = 0.477$; 0.16 at $\sigma = 0.30$; 0.30 at $\sigma = 0.60$	Main sec:selection-results	price_equation.collective_optimum_fraction
$p^*(\sigma, F)$	Equilibrium builder frequency reached from rarity	[0, 1]	1 for $F \leq 0.35$; interior otherwise	Supp. sec:si-intensity	price_equation.equilibrium_intensity, equilibrium_intensity_roots
Cov_G, Cov_g	Between- and within-group covariances of the Price partition	fraction	e.g. -0.009, +0.036 at $\sigma = 0.477$ (uniform ensemble)	Main eq:price; Supp. tab:partition-ensembles	price_equation.price_partition
$\bar{w}$	Population mean fitness in the Price identity	fraction	-	Main eq:price	same
$w_{\text{MB}}, w_{\text{NB}}$	Group-mean fitness of an all-building and an all-non-building group	fraction	-	Supp. eq:wMB, eq:wNB	price_equation.group_fitness_signaler, group_fitness_nonsignaler
$\Delta w(\sigma), \Delta w(\phi)$	Group-level advantage $w_{\text{MB}} - w_{\text{NB}}$ (distinguished by argument from the Layer-1 diagnostic $\Delta w(q)$)	fraction	-	Main fig:sigma_star; sec:bistable-emergence	price_equation.fitness_advantage, between_group_selection; emergence.rare_builder_fitness_advantage
ω	Convex-combination weight between multiplicative (0) and additive (1) fitness	[0, 1]	0 (primary); $\sigma^* \approx 0.53$ at $\omega = 1$	Supp. eq:convex	mode="convex", omega in critical_threshold_sigma_star; sigma_star_vs_omega
σ^*	Maintenance (critical) threshold at which $w_{\text{MB}} = w_{\text{NB}}$	[0, 1]	≈ 0.477 at the $\delta = 1$ reference; ≈ 0.418 at $\delta = 0.10$; [0.24, 0.65] over the (C, n) rectangle; ≈ 0.43 central	Main tab:notation; Supp. tab:sigma	price_equation.sigma_star_self_consistent
σ^*_{base}		[0, 1]	≈ 0.498 at $C = 0.352$	Main sec:self-consistent-threshold	price_equation.initial_model_sigma_star; INITIAL_MODEL_PARAMS

Symbol	Meaning	Units / range	Value	Manuscript	Code
	Assumed-coefficient baseline $C/[\beta - (1 - C)\alpha]$ with $\alpha = 0.30$, $\beta = 0.90$				
α, β, r (assumed)	Stipulated coefficients of the assumed-coefficient baseline	fraction	0.30, 0.90, 0.75	Main sec:self-consistent-threshold	INITIAL_MODEL_PARAMS

1.7 Spread, spatial dynamics, and further predictions

Symbol	Meaning	Units / range	Value	Manuscript	Code
ϕ	Fraction of neighboring groups (or of a region) that build	[0, 1]	-	Main sec:bistable-emergence	frac_signalers argument
$k(M_g, \phi)$	Partner-availability degree $k_0 + \phi[k(M_g) - k_0]$	partners	-	Supp. eq:krare	emergence.rare_builder_fitness_advantage (k_rare)
$\sigma^*_{inv}(\phi)$	Invasion threshold for a rare builder among a fraction ϕ of building neighbors	[0, 1] or ∞	∞ at $\phi \leq 0.05$; 0.73 at 0.25; 0.59 at 0.5; 0.48 at 1	Supp. tab:sigma	emergence.sigma_star_invasion
$\phi^*(\sigma)$	Unstable saddle of the replicator dynamics	[0, 1]	≈ 0.29 at $\sigma = 0.70$	Main sec:bistable-emergence	emergence.phi_star
$\dot{\phi}$	Replicator equation $\phi(1 - \phi)[w_{MB}(\phi) - w_{NB}(\phi)]$	per unit time	-	Main sec:bistable-emergence	emergence.replicator_dynamics
$f(\phi)$	Reaction term $\phi(1 - \phi)[w_{MB}(\phi) - w_{NB}(\phi)]$	per unit time	-	Supp. eq:mean_field_reaction	spatial.reaction_term
$\int_0^1 f d\phi$	Reaction integral; sign sets the front direction	-	0 at σ_M	Supp. sec:si-spatial-derivations	spatial.reaction_integral
σ_M	Maxwell point at which the planar front is stationary	[0, 1]	≈ 0.60	Main tab:notation; Supp. tab:sigma	spatial.maxwell_point_sigma
D (diffusion)	Diffusion coefficient of the reaction-diffusion equation, $\phi h^2/4$ at lattice scale	lattice units ² per time	1 in all figures	Supp. eq:bistable_RD	D argument in spatial.solve_pde_2d
D(ϕ)	Degenerate state-dependent diffusivity of the asymmetric imitation	same	-	Supp. eq:phi_dependent_RD	spatial.degenerate_diffusion_table

Symbol	Meaning	Units / range	Value	Manuscript	Code
	rule, normalized to $D^{1/2}$ = D_0				
h, ρ (lattice)	Lattice spacing and flip-rate constant of the discrete model	lattice units, per time	$\tau\rho \leq 2$ in simulations	Supp. sec:si-continuum-limit	<code>spatial.stochastic_lattice_simulation</code>
$s_i, \phi_i^{\text{local}}$	Binary building state of lattice site i and its four-neighbor mean	$\{0, 1\}, [0, 1]$	-	Supp. sec:si-continuum-limit	same
μ	Spontaneous per-site origination probability	per group per step	metastable for $\mu \leq 10^{-2}$ at $\sigma = 0.55$	Supp. sec:si-continuum-limit	<code>spatial.origination_robustness_sweep</code>
$U(z)$	Heteroclinic front profile in the comoving frame	-	-	Supp. eq:variational_velocity	not implemented in closed form (velocity measured numerically)
v	Front velocity	lattice units per time	$\approx 0.234\sqrt{D}$ at $\sigma = 0.8$; sign change at σ_M	Supp. eq:variational_velocity	<code>spatial.traveling_wave_velocity_pde</code> , <code>traveling_wave_velocity_dimensional_estimate</code>
$W_M(\phi)$	Allen-Cahn potential $-\int_0^1 \phi f(s; \sigma_M) ds$ at the Maxwell point	-	-	Supp. eq:critical_nucleus	<code>spatial.potential_W</code>
σ_{int}	Coexistence interfacial tension $\sqrt{D} \int_0^1 \sqrt{2(W_M - W_{M,\text{min}})} d\phi$	-	-	Supp. eq:critical_nucleus	<code>spatial.interfacial_tension</code>
ΔE	Bulk tilt $-\int_0^1 f d\phi$ at the operating σ	-	-	Supp. eq:critical_nucleus	inside <code>critical_nucleus_radius_allen_cahn</code>
R_c, R_c^{AC}	Critical nucleus radius (PDE bisection) and thin-wall estimate $\sigma_{\text{int}}/ \Delta E $	lattice units	≈ 5.0 (constant D); ≈ 5.1 (non-divergence $D(\phi)$); ≈ 3.6 (divergence form); $R_c^{\text{AC}} \approx 3.7$ at $\sigma = 0.8$	Main sec:spatial; Supp. eq:critical_nucleus	<code>spatial.critical_nucleus_radius_pde</code> , <code>critical_nucleus_radius_pde_degenerate</code> , <code>critical_nucleus_radius_allen_cahn</code>
ℓ	Interface width $\sqrt{(D/ f'(\phi^*))}$	lattice units	≈ 4.3 at $\sigma = 0.8$	Supp. sec:si-spatial-derivations	<code>spatial.critical_nucleus_radius_dimensional_estimate</code>
S^*	Formal droplet barrier $\pi\sigma_{\text{int}}^2/ \Delta E $ (thin-interface limit; no rate reported)	-	-	Supp. eq:critical_action	<code>spatial.critical_droplet_action</code>
T_{eff}	Effective noise temperature of the unreported Kramers form	-	not supplied	Supp. sec:si-spatial-derivations	<code>spatial.nucleation_rate_estimate</code>
z, z_{max}		normalized	$z_{\text{max}} = 1$	Supp. sec:si-placement	<code>PLACEMENT["z_max"]</code>

Symbol	Meaning	Units / range	Value	Manuscript	Code
	Placement exposure coordinate from resource core (0) to periphery				
η_W, η_C, η_X	Visibility to within-group, competitive, and cooperative audiences: e^{-bz} , $z/(z_{1/2} + z)$	[0, 1]	$b = 3.0, z_{1/2} = 0.3$	Supp. sec:si-placement	placement.visibility_within, visibility_between; PLACEMENT["b"], PLACEMENT["z_half"]
$G(z)$	Resource and logistics cost γz^2 (placement γ)	payoff units	$\gamma = 1.0$	Supp. sec:si-placement	placement.resource_cost; PLACEMENT["gamma"]
R (placement)	Return scale multiplying the audience-weighted visibility	payoff units	1.0 (results R-independent)	Supp. eq:placement-payoff	PLACEMENT["R"]
$\Pi(z), z^*$	Placement payoff and its maximizer	payoff units, [0, $z_{\max}$]	$z^* \approx 0.018$ at the emergence operating point; 0 at the established-practice anchor	Supp. eq:placement-payoff, eq:placement-interior	placement.placement_payoff, optimal_placement, comparative_static
q_i, q_j (contest)	Capacities of two contesting groups; unit labor cost $c_i = 1/q_i$	payoff units	-	Supp. eq:csf-equilibrium	competition.dyadic_effort
V (prize)	Common contest prize	payoff units	1.0	Supp. sec:si-competition	COMPETITION["V"]
x_i^*, X^*	Directed and total equilibrium contest effort $V q_i^2 q_j / (q_i + q_j)^2$, $V q_i q_j / (q_i + q_j)$	payoff units	-	Supp. eq:csf-equilibrium	competition.dyadic_effort, dyadic_total
π_i	Contest-success share $x_i / (x_i + x_j) = q_i / (q_i + q_j)$ at equilibrium	[0, 1]	-	Supp. sec:si-competition	inside verify_dyadic_equilibrium
$\bar{q}, \delta_q$	Mean capacity and asymmetry $(q_i - q_j) / (q_i + q_j)$	payoff units, [-1, 1]	-	Supp. eq:csf-equilibrium	competition.dyadic_asymmetry
Gini	Gini coefficient of monument sizes (pushforward of capacity through x^*) vs of capacity	[0, 1]	0.47 $\rightarrow$ 0.49 for the illustrative sample	Supp. sec:si-size-inequality	size_inequality.gini, monument_sizes, predicted_size_gini

2. Equation-to-function crosswalk

Each labelled equation in the main text and the Supplementary Information is listed with the function(s) that implement or verify it. "Inline" means the expression is evaluated inside the named function without a separate entry point.

2.1 Main text

Label	Content	Implementation
eq:spence	$\partial^2 c / \partial x \partial q = -x / q^2 < 0$	layer1.quadratic_cost (cost and partials), layer1.verify_spence_condition (symbolic check, also for the power and exponential costs)
eq:xstar	$x^*(q) = \sqrt{(\lambda_W (q^2 - q_{\min}^2))}$	layer1.equilibrium_investment (numerical), layer1.derive_equilibrium (symbolic ODE solution); alternative cost and benefit forms via layer1.cost_function_sensitivity, benefit_function_sensitivity
eq:depreciation	$M_g(t+1) = (1 - \delta)M_g(t) + I_g(t)$	layer1.monument_stock_trajectory (trajectory), layer1.effective_monument_stock (steady state I_g/δ)
eq:survival	$S(\sigma, k) = 1 - \sigma / (1 + \gamma k)$	layer3.survival_probability; $\alpha(k)$ via layer3.vulnerability_coefficient
eq:price	$\bar{w} \Delta \bar{p} = \text{Cov}_G(W_g, p_g) + E_G[\text{Cov}_g(w_i, p_i)]$	price_equation.price_partition

Unlabelled quantities stated inline in the main text: the free-rider payoff $q + \lambda_W q_{\min}$ in the caption of fig:equilibrium (layer1.free_rider_deterrent, layer1.fitness_at_deviation); $\beta_0 = SK(s_W - C)$ in sec:within-group-differential (price_equation.within_group_regression); the steady state $M_g^* = I_g/\delta$ in sec:depreciation (layer1.effective_monument_stock). The closed forms that the EHS refit removed from the main text ($\sigma^*_{\text{base}} = C/[\beta - (1 - C)\alpha]$, $W_g = (1 - p_g C)SK$, the replicator equation, the time constant $\tau = -1/\ln(1 - \delta)$, the selection criterion $F\beta_1 + (1 - F)\beta_0$) are now stated in words there and given in the Supplementary Information; their implementations are price_equation.initial_model_sigma_star, price_equation.mixed_group_mean_fitness, emergence.replicator_dynamics, layer1.signal_half_life, and price_equation.emergence_criterion.

2.2 Supplementary Information

Label	Content	Implementation
eq:noise-decomp	$\sigma_0^2 = \sigma_{\text{deception}}^2 + \sigma_{\text{lag}}^2 + \sigma_{\text{mismatch}}^2$	layer2.constrained_sigma_0
eq:woa-deterrence	$\Pr[V > c(M_g + M_h)] = 1/(1 + \beta_d(M_g + M_h))$	layer2.war_of_attrition_deterrence; exponential-stake alternative layer2.mutual_assessment_conflict_prob_exponential_stake
eq:K-waravoidance	$K(\sigma) = 1 - W(\sigma)(1 - \phi r)$	inline in price_equation.group_fitness_signaler ($\phi = 1$), group_fitness_nonsignaler ($\phi = 0$), emergence.rare_builder_fitness_advantage (general ϕ)
eq:waravoid-advantage	$K_{\text{build}} - K_{\text{non}} = w_0 \sigma \phi r$	difference of the above; no standalone function
eq:positional-game	$b_i = \lambda_W(\hat{q}_i - \text{peer mean})$	mean-field transfers in price_equation.within_group_regression; the finite-n peer-excluded game is stated analytically and is not implemented
eq:sW	$s_W = \lambda_W(E[q] - q_{\min})$	price_equation.within_group_status_differential
eq:wS-si, eq:wN-si	Restatement of eq:full_fitness, eq:member_freerider in the free-rider section	as above

Label	Content	Implementation
eq:beta0-si	$\beta_0 = SK(s_W - C)$	price_equation.within_group_regression, within_group_selection (per unit SK); sampled form layer1.participation_differential_sampled
eq:emergence-criterion	$\text{sign}(\Delta\bar{p}) = \text{sign}(F\beta_1 + (1 - F)\beta_0)$	price_equation.emergence_criterion, with between_group_regression; roots and stability equilibrium_intensity_roots, from-rarity value equilibrium_intensity
eq:convex	$w(\omega) = (1 - \omega)w_{\text{mult}} + \omega w_{\text{add}}$	price_equation.critical_threshold_sigma_star(mode="convex", omega=...), sigma_star_self_consistent(mode="convex", omega=...), sweep sigma_star_vs_omega
eq:pconflict_self	Self-assessment conflict probability	layer2.self_assessment_conflict_prob, derived_conflict_reduction_self_assessment; Bourgeois alternative layer2.bourgeois_conflict_prob; all three surfaces layer2.compare_assessment_models
eq:mean_field_reaction	$d\phi/dt = \rho\phi(1 - \phi)\Delta w(\phi)$	spatial.reaction_term (ρ absorbed into time)
eq:phi_dependent_RD	$\partial_t \phi = f(\phi) + D(\phi)\nabla^2 \phi$ with degenerate $D(\phi)$	spatial.degenerate_diffusion_table, spatial.nondivergence_diffusion_2d (derived operator), spatial.divergence_diffusion_2d (conservative variant), spatial.solve_pde_2d_degenerate
eq:bistable_RD	$\partial_t \phi = \phi(1 - \phi)[w_{\text{MB}}(\phi) - w_{\text{NB}}(\phi)] + D\nabla^2 \phi$	spatial.solve_pde_2d, with spatial.laplacian_2d; lattice counterpart spatial.stochastic_lattice_simulation
eq:deltaw	$\Delta w(q) = \lambda_W q/2 + \lambda_W q_{\text{min}}^2/(2q)$	layer1.fitness_gain, layer1.derive_equilibrium_fitness
eq:Blambda	$B(\lambda_W) = E[\Delta w(q)]$ under uniform capacity	price_equation.average_signaling_benefit (closed form), average_signaling_benefit_numerical, layer1.expected_fitness_gain
eq:Cmodel	$C_{\text{model}}(\lambda_W) = E[c(x^*(q), q)]$	layer1.average_equilibrium_cost; inverse layer1.lambda_W_for_C; per-type cost layer1.cost_at_equilibrium
eq:placement-payoff	$\Pi(z) = R[\lambda_W \eta_W + \lambda_C \eta_C + \lambda_X \eta_X] - G(z)$	placement.placement_payoff, with visibility_within, visibility_between, resource_cost
eq:placement-interior	Core condition $R[\lambda_C \eta_C'(0) + \lambda_X \eta_X'(0)] > G'(0) + R\lambda_W \eta_W'(0) $	evaluated through placement.optimal_placement ($z^* > 0$ exactly when the condition and the endpoint conditions hold); comparative statics placement.comparative_static; no standalone predicate
eq:csf-equilibrium	$x_i^* = V q_i^2 q_j / (q_i + q_j)^2$, $X^* = V q_i q_j / (q_i + q_j)$	competition.dyadic_effort, dyadic_total, dyadic_asymmetry; symbolic check verify_dyadic_equilibrium; regional sum regional_investment
eq:variational_velocity	$v \int (dU/dz)^2 dz = \int_0^1 f d\phi$	velocity measured by spatial.traveling_wave_velocity_pde; sign and $\sqrt{D}$ scaling via spatial.reaction_integral and traveling_wave_velocity_dimensional_estimate; the identity itself is not evaluated because $U(z)$ has no closed form
eq:critical_nucleus	$R_c^{\text{AC}} = \sigma_{\text{int}}/ \Delta E $ with the coexistence tension	spatial.critical_nucleus_radius_allen_cahn, spatial.interfacial_tension, spatial.potential_W, spatial.maxwell_point_sigma; canonical PDE value spatial.critical_nucleus_radius_pde
eq:critical_action	$S^* = \pi \sigma_{\text{int}}^2 / \Delta E $	spatial.critical_droplet_action; the unreported rate form spatial.nucleation_rate_estimate
eq:freerider_deterrent	$\Delta_{\text{fr}}(q) = \lambda_W(q - q_{\text{min}})^2/(2q)$	layer1.free_rider_deterrent, layer1.participation_equilibrium
eq:multiperiod_spence	$\partial^2 c_{\text{maint}} / \partial M_g \partial q = -\delta^2 M_g / q^2$	layer1.maintenance_cost, layer1.verify_multi_period_spence_condition
eq:reno_ratio	$R(M_g) = \delta M_g / (I_g - \delta M_g)$	layer1.maintenance_to_new_ratio
eq:pconflict	Mutual-assessment conflict probability with absolute deterrence	layer2.mutual_assessment_conflict_prob; layer2.effective_noise (σ_{eff}); layer2.war_of_attrition_deterrence (deterrence factor); reduction r via layer2.derived_conflict_reduction
eq:kdegree		

Label	Content	Implementation
	$k(M_g) = k_0 + k_{\max} M_g / (M_{1/2} + M_g)$	layer3.network_degree; alternatives network_degree_exponential, network_degree_hill, network_degree_piecewise, calibrated by calibrate_alternative_network_forms
eq:lambdaX	$\lambda_X = (dk/dM_g)(\sigma\gamma/(1 + \gamma k)^2)$	layer3.compute_lambda_X, with layer3.network_degree_derivative
eq:full_fitness, eq:member_freerider	w_S, w_N with positional status	price_equation.within_group_regression (returns $w_S - w_N$), price_equation.mixed_group_mean_fitness (returns W_g); the pair is computed by a private helper the two public functions share
eq:wMB, eq:wNB	$w_{MB} = (1 - C)(1 - \alpha_{\text{eff}} \sigma)(1 - W(1 - r))$, $w_{NB} = (1 - \beta_{\text{eff}} \sigma)(1 - W)$	price_equation.group_fitness_signaler, group_fitness_nonsignaler; difference fitness_advantage; root critical_threshold_sigma_star and the canonical wrapper sigma_star_self_consistent
eq:krare	$k(M_g, \phi) = k_0 + \phi[k(M_g) - k_0]$	inline in emergence.rare_builder_fitness_advantage (k_{rare}), which also applies the ϕ -weighted war-avoidance factor

Unlabelled displayed quantities in the Supplementary Information: $\lambda_s = a_s^W \lambda_W + a_s^C \tilde{\lambda}_C + a_s^X \tilde{\lambda}_X$ and $\bar{V}_s = \lambda_s p_s \bar{A} - f_s$ (layer1.channel_effective_lambda, mean_payoff_multiplier, channel_dominance_condition, monument_dominance_threshold); the lattice transition probabilities $\tau p_i^{\text{local}} \max(0, \Delta w_i)$ and $\tau p(1 - \phi_i^{\text{local}}) \max(0, -\Delta w_i)$ (spatial.stochastic_lattice_simulation); the feedback map $\mathcal{T}(\lambda)$ and its Lipschitz bound (layer3.lambda_total_at_sigma, price_equation.lipschitz_derivative_analytical, stability_analysis); the pinning diagnostic (spatial.pinning_at_discontinuity); the alternative network forms (layer3.network_degree_exponential, network_degree_hill, network_degree_piecewise and their derivatives); the population-mean identity $E[\Delta_{\text{fr}}] = s_W - C_{\text{model}}$ (layer1.participation_differential_sampled); the σ_0 consistency check (layer2.constrained_sigma_0); the vulnerability recovery $\gamma = (1/\beta - 1)/k_0$ (layer3.calibrate_to_initial_vulnerability).

3. Function-to-equation crosswalk

Every public function and class in `src/signaling/` is listed by module with the equation or section it implements. Private helpers (leading underscore) are omitted.

3.1 layer1.py

Function	Implements
CostSpec, BenefitSpec	Containers for a cost $c(x, q)$ with partials and a benefit $b(\hat{q}, \lambda)$; Main Layer 1 primitives
SpenceResult, EquilibriumResult, FitnessResult	Result containers for the symbolic derivations of eq:spence, eq:xstar, and eq:deltaw
quadratic_cost	$c = x^2/(2q)$; eq:spence
power_cost, exponential_cost	Alternative cost forms; Supp. sec:si-functional-form
linear_benefit, concave_benefit	$b = \lambda \hat{q}$ and $b = \lambda \hat{q}^\gamma$; Supp. sec:si-functional-form
equilibrium_investment	eq:xstar
equilibrium_fitness	$w^*(q)$; Supp. sec:si-fitness-gain, Main fig:equilibrium
fitness_gain	eq:deltaw
cost_at_equilibrium	Per-type equilibrium cost $\lambda_W(q^2 - q_{\min}^2)/(2q)$; integrand of eq:Cmodel
average_equilibrium_cost	eq:Cmodel
lambda_W_for_C	Inverse of eq:Cmodel ($\lambda_W \approx 1.93 C_{\text{emp}}$); Supp. sec:si-Cmodel

Function	Implements
free_rider_deterrent	eq:freerider_deterrent
participation_equilibrium	Full-participation result and the μ_N sensitivity; Supp. sec:si-free-rider
participation_differential_sampled	$\beta_0/(SK) = \lambda_W E[(q - q_{\min})^2/(2q)]$ for an arbitrary sample; Supp. sec:si-free-rider
receiver_inference	$\hat{q}(x) = \sqrt{(x^2/\lambda_W + q_{\min}^2)}$, the schedule inversion; Main Layer 1
fitness_at_deviation	$w(x, q) = q - x^2/(2q) + \lambda_W \hat{q}(x)$; Supp. sec:si-incentive
pooling_fitness	Pooling comparison; Supp. sec:si-incentive
expected_fitness_gain	$E[\Delta w]$ under a specified capacity density; eq:Blambda and Supp. sec:si-capacity-dist
verify_spence_condition	Symbolic eq:spence for three cost forms
derive_equilibrium	Symbolic solution of $x \, dx/dq = \lambda_W q$ with $x(q_{\min}) = 0$; eq:xstar
derive_equilibrium_fitness	Symbolic $w^*(q)$ and $\Delta w(q)$; eq:deltaw
derive_aggregate_signal	$E[x^*]$, $E[M_g] = nE[x^*]$, $\text{Var}[M_g]$; Main "Group monument stock"
group_monument_stock	$I_g = \sum_i x^*(q_i)$; Main "Group monument stock"
expected_monument_stock	$I_g = nE[x^*(q)]$ under a chosen capacity distribution; anchor ≈ 10.3
effective_monument_stock	$M_g^* = I_g/\delta$; eq:depreciation steady state
monument_stock_trajectory	eq:depreciation
signal_half_life	$\ln 2/(-\ln(1 - \delta))$; companion to the time constant τ in Main sec:depreciation-results
maintenance_to_new_ratio	eq:reno_ratio
maintenance_cost	$c_{\text{maint}} = (\delta M_g)^2/(2q)$; eq:multiperiod_spence
verify_multi_period_spence_condition	eq:multiperiod_spence
cost_function_sensitivity, benefit_function_sensitivity	Numerical schedules under alternative forms; Supp. sec:si-functional-form, fig:functional-form
mean_payoff_multiplier	$\bar{A} \approx 0.533$; Supp. sec:si-channel
SignalChannel; MONUMENT_CHANNEL, FEAST_CHANNEL, RITUAL_CHANNEL, HUNTING_CHANNEL, VERBAL_CHANNEL, ALL_CHANNELS	Channel specifications (audience weights, fidelity p_s , fixed cost f_s); Supp. sec:si-channel-decomp
lambda_decomposition	$\lambda = \lambda_W + \tilde{\lambda}_C + \tilde{\lambda}_X$; Supp. sec:si-channel-decomp
channel_effective_lambda	λ_s ; Supp. sec:si-channel-decomp
signal_fidelity	p_s with windfall adjustment; Supp. sec:si-channel
channel_dominance_condition	Ranking by $\bar{V}_s = \lambda_s p_s \bar{A} - f_s$; Supp. sec:si-channel, fig:channel
monument_dominance_threshold	Monument-dominant region in $(\tilde{\lambda}_C, \tilde{\lambda}_X)$; Supp. sec:si-channel, sec:si-audience-weights
incremental_lambda	$\bar{V}_{\text{dominant}} - \bar{V}_{\text{second}}$; Supp. sec:si-channel

3.2 layer2.py

Function	Implements
effective_noise	$\sigma_{\text{eff}} = \sigma_0 / \sqrt{1 + \kappa(M_g + M_h)}$; Main sec:layer2
constrained_sigma_0	eq:noise-decomp
war_of_attrition_deterrence	eq:woa-deterrence
mutual_assessment_conflict_prob	eq:pconflict
mutual_assessment_conflict_prob_exponential_stake	eq:pconflict with $\exp(-\beta_d(M_g + M_h))$ deterrence; Supp. sec:si-waravoidance
mutual_assessment_exponential	Exponential approximation to the Gaussian assessment factor (code-side robustness variant; not reported in the manuscript)
self_assessment_conflict_prob	eq:pconflict_self
bourgeois_conflict_prob	Bourgeois convention; Supp. sec:si-self-assessment-sensitivity, fig:si-assessment-alternatives
war_of_attrition_cost	Escalation-cost remark; Supp. sec:si-woa
derived_conflict_reduction	$r = 1 - P_{\text{conflict}}/P_{\text{base}}$; Main sec:layer2, Supp. sec:si-assessment
derived_conflict_reduction_self_assessment	r_{self} ; Supp. sec:si-self-assessment-sensitivity
compute_lambda_C	$\lambda_C = -dL/dM$; Supp. sec:si-banach, Main tab:notation
compare_assessment_models	Conflict surfaces for the three regimes; Supp. fig:conflict, Supp. fig:si-assessment-alternatives
calibrate_conflict_reduction	Solves assessment parameters that reproduce a target r (comparison with the assumed 0.75); Supp. sec:si-assessment

3.3 layer3.py

Function	Implements
network_degree, network_degree_derivative	eq:kdegree and dk/dM_g
NetworkDegreeSpec	Bundle of an alternative $k(M)$ with its derivative; Supp. sec:si-network-robustness
network_degree_exponential(_derivative), network_degree_hill(_derivative), network_degree_pieewise(_derivative)	Alternative saturating forms; Supp. sec:si-network-robustness
calibrate_alternative_network_forms	Solves τ , K_H , and the piecewise slope to match the reference at M_e ; Supp. sec:si-network-robustness
survival_probability	eq:survival
vulnerability_coefficient	$\alpha(k) = 1/(1 + \gamma k)$; Main sec:layer3
derive_vulnerability_differential	α_{eff} , β_{eff} ; Main sec:layer3, Supp. sec:si-network
compute_lambda_X	eq:lambdaX
between_group_lambda_diagnostics	λ_C and λ_X evaluated at $M_g(\lambda_W)$ under the λ_W -only schedule; Main sec:cost-reward-consistency
lambda_total_at_sigma	

Function	Implements
	Fixed point of the composite-reward map $\mathcal{T}(\lambda)$ (robustness variant); Supp. sec:si-banach, sec:si-feedback
lambda_sigma_sweep	$\lambda(\sigma)$, M_g , k across σ for the variant; Supp. fig:feedback, fig:supp-feedback
form_network_signal_based, form_network_baseline	Agent-based signal-read and random-graph networks; Supp. sec:si-network, fig:si-network-calibration
calibrate_to_initial_vulnerability	γ that recovers $\beta = 0.90$ exactly (≈ 0.22); Supp. sec:si-network

3.4 price_equation.py

Function	Implements
individual_fitness	Illustrative composition of the layer factors through one member's fitness (diagnostic; not the objective generating eq:xstar)
average_signaling_benefit, average_signaling_benefit_numerical	eq:Blambda
within_group_fitness_factor, within_group_fitness_factor_numerical	$E_q[(1 - c^*(q))(1 + \lambda\hat{q})]$ diagnostic accompanying individual_fitness
group_fitness_signaler, group_fitness_nonsignaler	eq:wMB, eq:wNB (multiplicative, additive, and convex modes)
fitness_advantage	$\Delta w(\sigma) = w_{MB} - w_{NB}$; Main fig:sigma_starb
initial_model_sigma_star	$\sigma^*_{base} = C/[\beta - (1 - C)\alpha]$; Main sec:self-consistent-threshold
critical_threshold_sigma_star	Bracketed root-finding for σ^* ; modes multiplicative, additive, convex; off-locus $C_{exogenous}$; Main sec:self-consistent-threshold, Supp. sec:si-bivariate
sigma_star_self_consistent	Canonical threshold with $C = C_{model}(\lambda_W)$, derived r , α_{eff} ; optional delta for the depreciated stock; Main sec:results-sigma-star, sec:depreciation-results
lipschitz_derivative_analytical	Closed-form $\mathcal{T}'(\lambda)$; Supp. sec:si-banach
stability_analysis, stability_analysis_self_consistent	Threshold gradient, fixed-point uniqueness, and convergence checks at σ^* ; Supp. sec:si-banach
within_group_selection	$s_W - C$, the within-group term per unit SK; eq:beta0-si
between_group_selection	$w_{MB} - w_{NB}$ with exogenous C (two homogeneous groups); distinct from Cov_G
classify_equilibrium	Regime classification by σ against σ^* and channel dominance; Main sec:channel
phase_space	Δw over (σ, C) ; Supp. sec:si-phase-space, fig:phase_space
sigma_star_comparison	σ^*_{base} and framework σ^* across C ; Main fig:sigma_stara
sigma_star_vs_delta	$\sigma^*(\delta)$; Main sec:depreciation-results
sigma_star_vs_param	One-parameter sweep of σ^* ; Supp. sec:si-sensitivity, sec:si-k0
sigma_star_vs_omega	$\sigma^*(\omega)$; Supp. sec:si-convex, fig:convex
sigma_star_bivariate	σ^* on a two-parameter grid; Supp. sec:si-bivariate, sec:si-omega-k0-joint
sensitivity_tornado	One-at-a-time sensitivity; Supp. fig:sensitivityc

Function	Implements
within_group_status_differential	eq:sW
mixed_group_mean_fitness	$W_g(p_g) = (1 - p_g C)SK$; Main sec:within-between
within_group_regression	$\beta_0 = w_S - w_N$; eq:beta0-si (status_scale = 0 recovers the altruism case $-C SK$)
between_group_regression	$\beta_1 = dW_g/dp$; Supp. eq:emergence-criterion
emergence_criterion	$F\beta_1 + (1 - F)\beta_0$; eq:emergence-criterion
equilibrium_intensity_roots	All zeros of the selection gradient with stability; Supp. sec:si-intensity
equilibrium_intensity	$p^*(\sigma, F)$ from rarity; Supp. sec:si-intensity, fig:si-intensity
price_partition	eq:price over an explicit composition ensemble; Supp. tab:partition-ensembles
collective_optimum_fraction	p_{opt} ; Main sec:selection-results

3.5 emergence.py

Function	Implements
rare_builder_fitness_advantage	$w_{MB}(\phi) - w_{NB}(\phi)$ with eq:krare and the ϕ -weighted eq:K-waravoidance; Main sec:bistable-emergence, fig:emergencea
sigma_star_invasion	$\sigma^*_{inv}(\phi)$; Supp. tab:sigma, Supp. fig:supp-emergence-saddle
phi_star	Saddle $\phi^*(\sigma)$; Main fig:emergenceb
replicator_dynamics	$\dot{\phi} = \phi(1 - \phi)[w_{MB}(\phi) - w_{NB}(\phi)]$; Main sec:bistable-emergence

3.6 spatial.py

Function	Implements
reaction_term	$f(\phi)$; eq:mean_field_reaction, eq:bistable_RD
reaction_integral	$\int_0^1 f d\phi$; Supp. sec:si-spatial-derivations
maxwell_point_sigma	σ_M ; Supp. tab:sigma
laplacian_2d	Five-point ∇^2 ; eq:bistable_RD
divergence_diffusion_2d, nondivergence_diffusion_2d	$\nabla \cdot (D(\phi)\nabla\phi)$ and $D(\phi)\nabla^2\phi$; eq:phi_dependent_RD
PDEResult, solve_pde_2d	Explicit-Euler integration of eq:bistable_RD; Main fig:spatial-main
degenerate_diffusion_table, solve_pde_2d_degenerate	$D(\phi)$ of the asymmetric rule and its integration; eq:phi_dependent_RD
traveling_wave_velocity_dimensional_estimate	$\text{sign}(l)\sqrt{(2D l)}$ reference estimate; Supp. fig:spatial-suppb
traveling_wave_velocity_pde	Measured v ; Supp. eq:variational_velocity discussion
potential_W, interfacial_tension	$W_M(\phi)$ and σ_{int} ; eq:critical_nucleus
critical_nucleus_radius_allen_cahn	R_c^{AC} ; eq:critical_nucleus
critical_nucleus_radius_dimensional_estimate	$\ell = \sqrt{(D/ f'(\phi^*))}$; Supp. sec:si-spatial-derivations
critical_nucleus_radius_pde, critical_nucleus_radius_pde_degenerate	R_c by bisection under constant and degenerate D ; Main sec:spatial, Supp. sec:si-spatial-derivations

Function	Implements
critical_droplet_action	eq:critical_action
nucleation_rate_estimate	Kramers form $A \exp(-S^*/T_{\text{eff}})$ (no value reported in the manuscript)
pinning_at_discontinuity	Front pinning at a σ step; Main fig:spatial-mainb
stochastic_lattice_simulation	Discrete imitation model; Supp. sec:si-continuum-limit
origination_robustness_sweep	Origination-flux check; Supp. sec:si-continuum-limit

3.7 placement.py, competition.py, size_inequality.py

Function	Implements
placement.visibility_between, visibility_within, resource_cost	$\eta_C = \eta_X = z/(z_{1/2} + z)$, $\eta_W = e^{-bz}$, $G = \gamma z^2$; Supp. sec:si-placement
placement.placement_payoff	eq:placement-payoff
placement.optimal_placement	$z^* = \arg \max \Pi$; interiority per eq:placement-interior
placement.comparative_static	$dz^*/d\lambda_C > 0$, $dz^*/d\lambda_X > 0$, $dz^*/d\gamma < 0$; Supp. fig:placementb
competition.dyadic_effort, dyadic_total, dyadic_asymmetry	eq:csf-equilibrium
competition.verify_dyadic_equilibrium	Symbolic residual check of eq:csf-equilibrium
competition.regional_investment	Σ over neighbor pairs of X^* ; Supp. sec:si-competition regional corollary
size_inequality.gini, monument_sizes, predicted_size_gini	Pushforward of capacity through x^* ; Supp. sec:si-size-inequality

3.8 calibration.py

Module-level constants, grouped by the symbol they set (values in Section 1): DEFAULT_Q_MIN, DEFAULT_Q_MAX ($q_{\min}$, $q_{\max}$); DEFAULT_N_COMPUTE and alias DEFAULT_N (n); DEFAULT_LAMBDA (exploration λ , 0.5); DEFAULT_LAMBDA_W_ANCHOR ($\lambda_W = 0.68$); DEFAULT_LAM_W, DEFAULT_LAM_C, DEFAULT_LAM_X (emergence-stage λ_W , $\tilde{\lambda}_C$, $\tilde{\lambda}_X$); DEFAULT_DELTA, DELTA_SENSITIVITY_RANGE (δ); DEFAULT_WAR_COST (w_0); DEFAULT_RHO * (ρ_s), DEFAULT_WINDFALL_PROB, DEFAULT_FIXED_COST * (f_s); DEFAULT_SIGMA_Q, DEFAULT_RHO_PROD_FIGHT, DEFAULT_VAR_RHP (noise decomposition); DEFAULT_SIGMA_0, DEFAULT_KAPPA, DEFAULT_T_0, DEFAULT_BETA_CONFLICT, DEFAULT_V, DEFAULT_D, DEFAULT_P_BASE, DEFAULT_ALPHA_CONFLICT, DEFAULT_M_THRESH, DEFAULT_M_SCALE, DEFAULT_M_OWN, DEFAULT_STEEPNESS, DEFAULT_C_RATE (Layer 2); DEFAULT_GAMMA, DEFAULT_K_0, DEFAULT_K_MAX, DEFAULT_M_HALF, DEFAULT_P_CONNECT_BASE (Layer 3); DEFAULT_CONFLICT_MORTALITY, DEFAULT_CONFLICT_COST, DEFAULT_DISPUTE_FREQ, DEFAULT_N_NEIGHBORS (λ_C diagnostics and the assumed-coefficient baseline); PLACEMENT, COMPETITION (placement and contest parameters). The dictionaries RAPA_NUI, RAPA_ITI, INITIAL_MODEL_PARAMS, and CALIBRATION_TARGETS hold the case-study cost anchor $C = 0.35$, the assumed-coefficient baseline ($\alpha = 0.30$, $\beta = 0.90$, $r = 0.75$), and the published calibration ranges; only RAPA_NUI["C"] and INITIAL_MODEL_PARAMS are consumed by the manuscript's computations.

3.9 plotting.py

Figure utilities that consume the functions above and accept an ax argument for composition: setup_paper_style; plot_investment_function, plot_fitness_landscape, plot_fitness_gain (Main fig:equilibrium, Supp. fig:functional-form, fig:incentive); plot_channel_comparison, plot_dominance_region, plot_fidelity_sensitivity (Supp. fig:channel, fig:supp-audience-weights); plot_conflict_probability_surface, plot_assessment_model_comparison (Supp. fig:conflict, Supp. fig:si-assessment-alternatives); plot_network_degree, plot_survival_surface, plot_vulnerability_differential, plot_lambda_sigma_feedback (Supp. fig:network, fig:feedback); plot_sigma_star_comparison, plot_phase_space (Main fig:sigma_star, Supp. fig:phase_space). The palette constants PAPER_COLORS, PALETTE, SEQUENTIAL_CMAP, DIVERGING_CMAP, and TWO_CLASS_COLORS set the colorblind-safe scheme used throughout.

4. Entries not verified against both sources

Every table row above was checked against both the manuscript and the code. Three correspondences are partial rather than one-to-one and are recorded here so a reader does not look for a function that does not exist: the finite-n peer-excluded positional game of Supp. eq:positional-game is stated analytically and enters the code only through its mean-field transfers; the variational velocity identity of Supp. eq:variational_velocity is not evaluated because the front profile has no closed form, and the velocity is measured from PDE integration instead; and the placement core condition of Supp. eq:placement-interior has no standalone predicate and is checked through the sign of the optimizer's output.